

# KALLE HEINONEN

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## PERSONAL DETAILS

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I am a 25 yearold MSc in mathematics. I graduated from the study track of geometry, algebra and topology at the University of Helsinki. My master's thesis was on the Poincaré-Hopf index theorem, which relates data from isolated singularities of vector fields on closed manifolds to it's purely topological invariants. Specifcly a proof utilizing de Rham theory and Morse theory.

## EDUCATION

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**Bachelor of Science in Mathematics**, University of Helsinki Sep 2020 - May 2023

Minor in computer science

**Master of Science in Mathematics**, University of Helsinki Sep 2023 - April 2025

## SKILLS

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|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Technical Skills</b>           | Problem solving, LaTeX, Python, Haskell, basics of Matlab, R, Excel and SQL.                                                                         |
| <b>Mathematics knowledge</b>      | <a href="https://kh-xyz.github.io/mathlist">https://kh-xyz.github.io/mathlist</a> ← elaborated here                                                  |
| <b>Computer science knowledge</b> | The basic minor studies such as python programming, computer organization, algorithms and databases. Additionally functional programming in Haskell. |
| <b>Languages</b>                  | Finnish: Native, English: Full professional proficiency, Swedish: ~B1.                                                                               |

## JOB EXPERIENCE

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| - Non-military service at Aalto univerity Department of Mathematics and System's analysis | Sep 2025- Aug 2026  |
| - Live musician - Freelance                                                               | Sep 2021 - Current  |
| - High school mathmematics / computer science substitute teacher - Freelance              | Sep 2021 - Mar 2025 |
| - Teaching assistant at University of Helsinki & Aalto university - Seasonal              | Oct 2021 - May 2026 |
| - Villa host at Hämäläisten Ylioppilassäätiö - Full-time                                  | Aug 2026            |

## OTHER RESPONSIBILITIES

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| - Board member and other responsibilities at Matrix ry                                                                              | Jan 2021 - May 2025 |
| - Vice chair and other activities at Moodi ry                                                                                       | Jan 2024 - Dec 2026 |
| - Board and other activities at Scout organization Vuosaaren Vesipääskyt ry                                                         | Sep 2008 - Mar 2024 |
| - Student member of the board of the Master's program in mathematics and statistics and Bachelor's program in mathematical sciences | Jan 2023 - Jun 2025 |
| - Multiple responsibilities at Hämäläis-osakunta student nation                                                                     | Jan 2021 - Dec 2025 |
| - Vice representative in the representative council at Helsingin yliopiston ylioppilaskunta                                         | Jan 2023 - Dec 2024 |
| - Vice artistic director at Polirytmii ry                                                                                           | Jan 2026 - Dec 2026 |

## REFERENCES

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You may ask for letters of recommandation from my thesis advisor professor Pekka Pankka [pekka.pankka@helsinki.fi](mailto:pekka.pankka@helsinki.fi), or secondary thesis inspector university lecturer Erik Elfving [erik.elfving@helsinki.fi](mailto:erik.elfving@helsinki.fi).